

```

module traffic_light_controller_tb;

    reg clk;

    reg rst_n;

    reg timer_done;

    wire red;

    wire grn;

    wire yel;

    traffic_light_controller uut (

        .clk(clk),

        .rst_n(rst_n),

        .timer_done(timer_done),

        .red(red),

        .grn(grn),

        .yel(yel)

    );

    initial begin

        clk = 0;

        forever #5 clk = ~clk;

    end

    initial begin

        rst_n = 0;

        timer_done = 0;

        #20;

        rst_n = 1;

        // RED -> GREEN

        #20 timer_done = 1;

        #10 timer_done = 0;

```

```

// GREEN -> YELLOW
#20 timer_done = 1;
#10 timer_done = 0;
// YELLOW -> RED
#20 timer_done = 1;
#10 timer_done = 0;
// Repeat once more
#20 timer_done = 1;
#10 timer_done = 0;
#50;
$finish;
end

// Monitor Signals
initial begin
    $monitor("Time=%0t rst_n=%b timer_done=%b RED=%b GREEN=%b YELLOW=%b",
        $time, rst_n, timer_done, red, grn, yel);
end

// Generate VCD file for GTKWave
initial begin
    $dumpfile("traffic_light.vcd");
    $dumpvars(0, traffic_light_controller_tb);
end

endmodule

```